

$$1) G_c = \frac{\pi \sigma^2 a}{E} k$$

$$\frac{G_c E}{\pi} = \sigma^2 a \quad \therefore k=1$$

$$= (480 \times 10^6)^2 \left(\frac{40}{2} \times 10^{-3} \right)$$

$$= 4.608 \times 10^{15}$$

$$\frac{G_c E}{\pi} = \sigma^2 \left(\frac{100}{2} \times 10^{-3} \right)$$

$$4.608 \times 10^{15} = 50 \times 10^{-3} \sigma^2$$

$$\sigma = 303.5786554 \times 10^6 \text{ Pa}$$

$$\approx 303.6 \text{ MPa}$$

$$2) G_c = \frac{\sigma^2 a \pi}{E} k$$

Plane stress, so $k=1$

$$0.01 \times 10^3 = \frac{(3600 \times 10^6)^2 a \pi}{69 \times 10^9}$$

$$a = 1.694705413 \times 10^{-8} \text{ m}$$

$$2a = 3.389410825 \times 10^{-8} \text{ m}$$

$$\approx 3.38941 \times 10^{-8} \text{ m}$$

If the defect size is greater than this value, then the glass plate will fracture at a lower tensile stress value than 3600 MPa. In other words, the experimentally observed strength will be less than the theoretical tensile strength.

$$3) G_c = \frac{\sigma^2 a \pi}{E} k \quad K = \sigma \sqrt{\pi a}$$

$$\frac{G_c E}{k} = \sigma^2 a \pi$$

$$\sqrt{\frac{G_c E}{k}} = \sigma \sqrt{\pi a}$$

$$K = \sqrt{\frac{G_c E}{k}}$$

Assuming plane stress,

$$K = 1$$

$$K_c = \sqrt{\frac{0.01 \times 10^3 \times 69 \times 10^9}{1}}$$

$$= 830.6623863 \times 10^3 \text{ Pa m}^{\frac{1}{2}}$$

$$K = \sigma \sqrt{\pi a}$$

$$= 500 \times 10^6 \sqrt{\pi (0.002 \times 10^{-3} \times \frac{1}{2})}$$

$$= 886.2269255 \times 10^6 \text{ Pa m}^{\frac{1}{2}} > K_c$$

Since $K > K_c$, the crack will lead to fracture.

$$4) \frac{K_{finite} - K_{infinite}}{K_{infinite}} \times 100 \leq 1$$

$$\frac{\cancel{\sigma(\pi a)^{\frac{1}{2}}} Y - \cancel{\sigma(\pi a)^{\frac{1}{2}}}}{\cancel{\sigma(\pi a)^{\frac{1}{2}}}} \leq 0.01$$

$$\frac{Y-1}{1} \leq 0.01$$

$$Y \leq 1.01$$

$$\left(\frac{W}{\pi a} \tan \frac{\pi a}{W} \right)^{\frac{1}{2}} \leq 1.01$$

$$\tan \frac{\pi a}{W} \leq \frac{\pi a}{W} (1.01)^2$$

$$\text{Let } x = \frac{a}{W},$$

$$\tan \pi x \leq (1.01)^2 \pi x$$

Solving the equation,

$$x \leq 0.0772381604$$

$$\therefore \frac{a}{W} \leq 0.0772381604$$

$$a \leq 0.0772W$$

4) Hence, if $a \leq 0.0772W$, the difference will be less than 1%.

In other words, if $2a < 0.154W$, the difference will be less than 1%.

Thus, if the crack length is less than 15.4% of the width of the plate, the difference will be less than 1%.

For crack lengths less than 15.4% of the width of the plate, the dimensions of the plate do not affect the stress intensity much, and the infinite plate assumption will still give result.

5) For the initial slope, use point (0.34, 16.15).

95% of the slope is:

$$0.95 \left(\frac{17}{0.34} \right) = \frac{16.15}{0.34}$$

The 95% slope line intersects the load displacement curve at 19 kN. Hence, we take $F_Q = 19 \text{ kN}$.

From the graph, $F_{\max} = 20 \text{ kN}$. For the test to be valid, $\frac{F_{\max}}{F_Q} < 1.1$.

$$\frac{20}{19} \approx 1.053 < 1.1$$

$\therefore$ The test is valid.

$$K_Q = \frac{F_Q}{BW^{\frac{1}{2}}} f_2 \left(\frac{a}{w} \right)$$

$$\begin{aligned} \frac{a}{w} &= \frac{25}{2 \times 25} \\ &= \frac{1}{2} \end{aligned}$$

5) From Table 3.1,

$$f_2\left(\frac{a}{w}\right) = 9.6$$

$$\therefore K_Q = \frac{F_Q}{B W^{\frac{1}{2}}} (9.6)$$

$$= \frac{19 \times 10^3}{25 \times 10^{-3} (25 \times 2 \times 10^{-3})^{\frac{1}{2}}} (9.6)$$

$$= 32.6287039 \text{ MPa m}^{\frac{1}{2}}$$

$$\approx 32.6 \text{ MPa m}^{\frac{1}{2}}$$

To determine whether $K_Q = K_{Ic}$

$$B, a, (W-a) \geq 2.5 \left(\frac{K_Q}{\sigma_Y} \right)^2$$

$$0.45 < \frac{a}{w} < 0.55$$

$$2.5 \left(\frac{K_Q}{\sigma_Y} \right)^2 = 2.5 \left(\frac{32.6 \times 10^6}{650 \times 10^6} \right)^2$$

$$= 6.2996 \text{ mm}$$

$$\approx 6.3 \text{ mm}$$

$$B = 25 \text{ mm} > 6.3 \text{ mm} \quad a = 25 \text{ mm} > 6.3 \text{ mm}$$

$$W-a = 50-25 = 25 \text{ mm} > 6.3 \text{ mm}$$

$$\frac{a}{w} = 0.5 > 0.45 \text{ and } < 0.55$$

$$\therefore K_Q = K_{Ic} = 32.6 \text{ MPa m}^{\frac{1}{2}}$$